SOKOINE UNIVERSITY OF AGRICULTURE
School of Engineering and Technology

Course: ABE 721 - Agricultural Systems Modelling and Simulation
Instructor: Kadeghe Fue, PhD, P.Eng(T)

ASSIGNMENT 2
Module 2: Models of Systems

Instructions:

- This assignment bridges the gap between conceptual system boundaries (Module 1) and explicit mathematical formulation.
- Carefully read the ”Significance” statement for each question to understand the practical engineering application of the theory.
- Show all logical steps and define all variables clearly when writing differential equations.

Tier 1: Simple (Conceptual Foundations)

Question 1.1: Empirical vs. Mechanistic Architectures

Significance of Question: *Knowing exactly when to deploy a data-driven empirical model versus a physics-based mechanistic model prevents catastrophic simulation failures under novel climate change scenarios.*

Prompt: Briefly compare and contrast Empirical models and Mechanistic models. Provide one specific agricultural engineering scenario where an Empirical model (like a neural network or regression) is highly appropriate, and one scenario where a Mechanistic model (using laws of physics) is strictly required.

Question 1.2: Spatial Domains

Significance of Question: *Spatial assumptions dictate both computational cost and accuracy; choosing an overly simplified spatial domain can hide critical localized failures, such as isolated groundwater contamination.*

Prompt: Define Lumped (ODE) models and Distributed (PDE) models. Why would a regional watershed nitrate leaching model require a Distributed approach, while predicting the internal air temperature of a 10m x 10m greenhouse might only require a Lumped approach?

Tier 2: Intermediate (Application and Logic)

Question 2.1: The Iterative Modelling Process

Significance of Question: *Recognizing that systems modelling is a highly iterative loop rather than a linear checklist prevents engineers from deploying structurally flawed simulations into the field.*

Prompt: The systems modelling process consists of several steps (System Definition, Conceptualization, Mathematical Formulation, Implementation, Calibration, Validation). Suppose you build a crop growth model, but during Step 6 (Statistical Validation), the model generates an unacceptably high RMSE compared to the real-world field data. Identify and explain the *two* preceding steps in the modelling cycle you must revisit to diagnose and fix the error.

Question 2.2: State-Space Variable Identification

Significance of Question: *Translating physical farm components into strict vector notation (**x** and **u**) is the fundamental prerequisite for designing automated digital control algorithms.*

Prompt: Consider a continuous-flow grain drying system. Wet grain enters the dryer at a specific moisture content, hot air is blown through it, and dry grain exits. Identify the following mathematical components for this system:

a) Define the primary State Variable ($\mathbf{x}(t)$) of the grain inside the dryer.

b) Define two Exogenous Input Variables ($\mathbf{u}(t)$) that force the system.

c) Define one fixed Parameter (**p**) relevant to the drying physics.

Tier 3: Hard (Synthesis and Formulation)

Question 3.1: Greenhouse Thermodynamics Formulation

Significance of Question: *Formulating the First Law of Thermodynamics into a strict Ordinary Differential Equation (ODE) allows engineers to design intelligent, energy-saving climate control algorithms for Controlled Environment Agriculture (CEA).*

Prompt: You are designing a lumped thermodynamic model for a greenhouse to predict the internal air temperature $T_{in}(t)$. The fundamental energy balance is:

$$C_{air}\frac{dT_{in}}{dt} = Q_{solar} + Q_{heater} - Q_{conduction} - Q_{ventilation}$$

Write out the expanded mathematical sub-equations for Q_{solar}, $Q_{conduction}$, and $Q_{ventilation}$. Define every parameter you use (e.g., transmissivity τ, mass flow rate $\dot{m}_{air}$). Finally, combine them to show the complete, expanded ODE for $\frac{dT_{in}}{dt}$.

Question 3.2: Tractor Draft Force and Fuel Coupling

Significance of Question: *Mathematically coupling implement soil mechanics with engine thermodynamics allows for the computational optimization of fuel consumption, directly impacting the financial viability of large-scale commercial farming.*

Prompt: The ASABE standard formulation for implement draft force (D) is a quadratic equation based on tractor operational speed (S):

$$D = F_i \times \left[A + B(S) + C(S)^2\right] \times W \times T$$

Assume you have calculated the exact Draft force (D in Newtons) for a specific plowing operation. Write a theoretical, logical explanation of how you would mathematically link this mechanical draft equation to a dynamic state-space ODE representing the transient volume of diesel fuel remaining in the tractor's tank: $\frac{dV_{fuel}}{dt}$. What intermediary physical conversions (e.g., Power, Engine Efficiency) are required?

Please submit your work to my email kadeghe.fue@sua.ac.tz by one week after completion of Module 2. Do not send this work to kadefue@sua.ac.tz.